VELOOP REWARDS - Data Dictionary

Data Analyst Internship, Task 01 | Work Package G

Authenticity notice: Every field below is **Proposed** or **To Be Verified**. Example values are illustrative and are not VELOOP data. No simulated or real VELOOP statistics appear here.

1. Summary

- **128 fields** across 10 entities (brief minimum: 75). 99 unique field names; the rest are repeated keys such as user_id that each table needs for joins.
- **Status:** 84 Proposed, 44 To Be Verified, 0 Confirmed (nothing has been verified by engineering).
- **Sensitivity:** 3 Public, 119 Internal, 6 Restricted. No personal-data fields exist.
- Fields were added only because a specification, event, KPI or table needs them. Nothing was added to reach a count.
- task_type and feature live on Tasks and are reached through task_id; event payloads still carry them (Event Tracking Plan).

2. Entities

Entity	Grain	Primary Key	Scope	Fields
Users	One row per user account	user_id	Core	21
Account_Status_History	One row per account status change	status_history_id	Core	7
Tasks	One row per task in the catalogue	task_id	Core	8
Task_Activity	One row per task attempt	task_activity_id	Core	17
Rewards	One row per reward	reward_id	Core	14
Wallet_Transactions	One row per ledger entry	transaction_id	Core	15
Withdrawals	One row per withdrawal request	withdrawal_id	Core	15
Referrals	One row per referral relationship	referral_id	Core	13
Feature_Events	One row per tracked event (raw event layer)	event_id	Core	12
Support_Tickets	One row per support request	ticket_id	Optional (support KPIs)	6

3. Sensitivity scheme

Level	Meaning	Access
Public	Already visible to users in the app, such as task titles	Open to any reader of the dictionary
Internal	Pseudonymous IDs, timestamps, statuses and categories	Analysts, engineering, product and operations; no personal data
Restricted	Money amounts, withdrawal methods and payout references	Named roles only; aggregated for wider reporting

4. Dictionary

Entity	Field Name	Data Type	Definition	Example	Required	Allowed Values	Validation	Source	Sensitivity	Analytical Use	Status
Users	user_id	string	Permanent anonymous key of the account	usr_example_0001	Yes	-	Non-null; unique; immutable	Backend	Internal	Join key to every other entity	To Be Verified
Users	registered_at	timestamp (UTC)	When the account was created	2025-01-15T10:30:00Z	Yes	-	ISO 8601 UTC; not in the future	Backend	Internal	Cohorts and growth	To Be Verified
Users	registration_source	enum	Channel through which the user arrived	organic	Yes	organic, referral, campaign, store, unknown (list to be agreed)	In approved list; 'unknown' instead of null	Frontend / Backend	Internal	Acquisition analysis	Proposed
Users	registration_method	enum	Sign-up route, category only (identity value never stored)	social	No	email, phone, social, other (to be agreed)	In approved list	Backend	Internal	Compare sign-up routes	Proposed
Users	platform	enum	Platform used at registration	android	Yes	android, ios, web (to be agreed)	In approved list	Frontend	Internal	Platform split	Proposed
Users	app_version	string	App version at registration	1.4.2	Yes	-	Matches version pattern	Frontend	Internal	Version-specific problems	Proposed
Users	country_code	string	Country of the account, country level only	IN	No	ISO 3166-1 alpha-2	Valid code; never precise location	Backend (derived)	Internal	Geographic distribution	Proposed
Users	language_code	string	Interface language	en	No	ISO 639-1	Valid code	Frontend	Internal	Content and support planning	Proposed
Users	referral_code_used	string	Referral code entered at sign-up	REF_EXAMPLE1	No	-	Matches an existing referral code or null	Backend	Internal	Link user to a referral	To Be Verified
Users	campaign_id	string	Marketing campaign of arrival	cmp_example_01	No	Approved campaign list	In list or null	Frontend / Backend	Internal	Campaign attribution	Proposed
Users	onboarding_status	enum	Current onboarding state	completed	Yes	not_started, in_progress, completed, skipped	In approved list	Backend	Internal	Onboarding funnel	Proposed
Users	onboarding_started_at	timestamp (UTC)	When onboarding began	2025-01-15T10:31:00Z	Conditional	-	Not before registered_at	Frontend	Internal	Time to start onboarding	Proposed
Users	onboarding_completed_at	timestamp (UTC)	When onboarding finished	2025-01-15T10:36:00Z	Conditional	-	Required when status = completed	Backend	Internal	Onboarding duration	Proposed
Users	first_activity_at	timestamp (UTC)	First qualifying activity	2025-01-15T11:00:00Z	Derived	-	Min of qualifying events; not before registered_at	Analytics layer	Internal	Time to first activity	Proposed
Users	first_reward_credited_at	timestamp (UTC)	First credited reward	2025-01-15T11:05:00Z	Derived	-	From first reward_credited	Analytics layer	Internal	Early value milestone	Proposed
Users	last_login_at	timestamp (UTC)	Latest login	2025-03-02T08:00:00Z	Derived	-	Max login time per user	Analytics layer	Internal	Return behaviour	Proposed
Users	last_activity_at	timestamp (UTC)	Latest qualifying activity	2025-03-02T08:20:00Z	Derived	-	Max qualifying event time	Analytics layer	Internal	Active vs inactive	Proposed
Users	account_status	enum	Current account state	active	Yes	active, restricted, closed (map to real statuses)	In approved list	Backend	Internal	Account state mix	To Be Verified
Users	status_changed_at	timestamp (UTC)	Time of latest status change	2025-02-10T09:00:00Z	Conditional	-	Required after any change	Backend	Internal	Status timing	Proposed
Users	account_closed_at	timestamp (UTC)	When the account closed	2025-04-01T00:00:00Z	Conditional	-	Required when status = closed	Backend	Internal	Churn timing	Proposed
Users	lifecycle_stage	enum	Single label for the user's stage	active	Derived	registered, onboarding, activated, active, inactive, restricted, closed	Computed from lifecycle rules	Analytics layer	Internal	Dashboard segmentation	Proposed
Account_Status_History	status_history_id	string	Key of one status change	sth_example_01	Yes	-	Unique	Backend	Internal	Audit of status changes	Proposed
Account_Status_History	user_id	string	Account that changed	usr_example_0001	Yes	-	Must exist in Users	Backend	Internal	Join to Users	To Be Verified
Account_Status_History	previous_status	enum	Status before the change	active	Yes	Same list as account_status	Differs from new_status	Backend	Internal	Read transitions	Proposed
Account_Status_History	new_status	enum	Status after the change	restricted	Yes	Same list as account_status	In approved list	Backend	Internal	Read transitions	Proposed
Account_Status_History	changed_at	timestamp (UTC)	When the change happened	2025-02-10T09:00:00Z	Yes	-	Not before registered_at	Backend	Internal	Timing of changes	Proposed

Entity	Field Name	Data Type	Definition	Example	Required	Allowed Values	Validation	Source	Sensitivity	Analytical Use	Status
Account_Status_History	status_change_actor	enum	Who caused the change	operations	Yes	user, system, operations	In approved list	Backend	Internal	Separate user vs operational changes	Proposed
Account_Status_History	status_reason_category	enum	Reason at category level, no free text	other	No	To be agreed with Operations	In list; no personal data	Backend	Internal	Explain changes	To Be Verified
Tasks	task_id	string	Key of a task in the catalogue (not an attempt)	task_example_01	Yes	-	Unique; immutable	Backend	Internal	Join key for task activity	To Be Verified
Tasks	task_type	enum	Category of task	video	Yes	Real VELOOP task types	In approved list	Backend	Public	Segment by type	To Be Verified
Tasks	feature	enum	Feature offering the task	watch_ads	Yes	Real VELOOP features	In approved list	Backend	Public	Segment by feature	To Be Verified
Tasks	task_title	string	Display name of the task	Example task	No	-	No personal data	Backend	Public	Readable reports	Proposed
Tasks	is_active	boolean	Whether the task is currently offered	true	Yes	true, false	Boolean	Backend	Internal	Exclude retired tasks	Proposed
Tasks	validation_method	enum	How completion is checked	provider	Yes	none, internal, provider	In approved list	Backend	Internal	Interpret validation results	To Be Verified
Tasks	provider_name	string	External provider, if any	provider_example	No	Approved provider list	Required when validation_method = provider	Backend	Internal	Provider reliability	To Be Verified
Tasks	created_at	timestamp (UTC)	When the task entered the catalogue	2025-01-01T00:00:00Z	Yes	-	ISO 8601 UTC	Backend	Internal	Catalogue history	Proposed
Task_Activity	task_activity_id	string	Key of one attempt by one user	ta_example_0001	Yes	-	Unique; immutable	Backend	Internal	Grain of the table	Proposed
Task_Activity	task_id	string	Task attempted (type and feature come from Tasks)	task_example_01	Yes	-	Must exist in Tasks	Backend	Internal	Join to Tasks	To Be Verified
Task_Activity	user_id	string	User who attempted	usr_example_0001	Yes	-	Must exist in Users	Backend	Internal	Join to Users	To Be Verified
Task_Activity	attempt_number	integer	Nth attempt by this user at this task	1	No	-	Integer >= 1	Backend	Internal	Repeat attempts	Proposed
Task_Activity	started_at	timestamp (UTC)	When the attempt began	2025-01-15T11:00:00Z	Yes	-	Not in the future	Backend	Internal	Task starts	Proposed
Task_Activity	completion_claimed_at	timestamp (UTC)	When completion was claimed	2025-01-15T11:02:00Z	Conditional	-	Not before started_at	Frontend / Provider	Internal	Claim-to-validation time	Proposed
Task_Activity	completed_at	timestamp (UTC)	When completion passed validation	2025-01-15T11:02:30Z	Conditional	-	Required when status = validated	Backend	Internal	Real completions	Proposed
Task_Activity	status	enum	State of the attempt	validated	Yes	started, completion_claimed, validated, failed, abandoned	In approved list	Backend	Internal	Funnel states	Proposed
Task_Activity	validation_status	enum	Result of validation	passed	Yes	not_required, pending, passed, failed	In approved list	Backend	Internal	Validation pass rate	Proposed
Task_Activity	validation_completed_at	timestamp (UTC)	When validation finished	2025-01-15T11:02:30Z	Conditional	-	Not before completion_claimed_at	Backend	Internal	Validation delay	Proposed
Task_Activity	reward_id	string	Reward linked to this attempt	rw_example_0001	Conditional	-	Must exist in Rewards	Backend	Internal	Link activity to reward	Proposed
Task_Activity	reward_amount	decimal	Reward amount tied to the attempt	10.00	Conditional	-	Not negative	Backend	Internal	Reward by task	Proposed
Task_Activity	reward_unit	enum	Unit of the reward	unit_example	Conditional	Real VELOOP units	Required when amount present	Backend	Internal	Prevents mixed-unit sums	To Be Verified
Task_Activity	failure_category	enum	Why the attempt failed	validation_failed	Conditional	validation_failed, tracking_failed, user_abandoned, system_error, provider_error, unknown	Required when failed or abandoned	Backend	Internal	Explain drop-off	Proposed
Task_Activity	failure_detail_code	string	VELOOP internal code, only if one exists	-	No	-	No personal data	Backend	Internal	Detail beyond category	To Be Verified
Task_Activity	source	enum	Where the record came from	backend	Yes	frontend, backend, provider	In approved list	Backend	Internal	Trust and reconciliation	Proposed
Task_Activity	provider_reference_id	string	Provider's own id	prov_example_01	No	-	Unique per provider	Provider	Internal	Match external records	To Be Verified
Rewards	reward_id	string	Key of one reward	rw_example_0001	Yes	-	Unique; immutable	Backend	Internal	Grain of the table	Proposed
Rewards	user_id	string	Reward owner	usr_example_0001	Yes	-	Must exist in Users	Backend	Internal	Join to Users	To Be Verified
Rewards	task_activity_id	string	Activity that earned it, if any	ta_example_0001	No	-	Must exist when present	Backend	Internal	Link to activity	Proposed
Rewards	source_feature	enum	Feature that produced the reward	spin	Yes	Real VELOOP features	In approved list	Backend	Internal	Reward activity by feature	Proposed

Entity	Field Name	Data Type	Definition	Example	Required	Allowed Values	Validation	Source	Sensitivity	Analytical Use	Status
Rewards	reward_amount	decimal	Reward amount	10.00	Yes	-	Greater than 0	Backend	Internal	Reward volume	Proposed
Rewards	reward_unit	enum	Unit of the reward	unit_example	Yes	Real VELOOP units	In approved list	Backend	Internal	Prevents mixed-unit sums	To Be Verified
Rewards	reward_status	enum	Current reward state	credited	Yes	pending, credited, reversed, rejected	In approved list	Backend	Internal	Reward state mix	Proposed
Rewards	pending_reason	enum	Why the reward is held	validation	Conditional	To be agreed	Required when pending	Backend	Internal	Pending queue	Proposed
Rewards	created_at	timestamp (UTC)	When the reward was created	2025-01-15T11:02:30 Z	Yes	-	Not in the future	Backend	Internal	Reward timing	Proposed
Rewards	credited_at	timestamp (UTC)	When it reached the wallet	2025-01-15T11:02:31 Z	Conditional	-	Required when credited	Backend	Internal	Credit delay	Proposed
Rewards	reversed_at	timestamp (UTC)	When a credit was reversed	2025-01-16T09:00:00 Z	Conditional	-	Required when reversed	Backend	Internal	Reversal timing	Proposed
Rewards	original_reward_id	string	Reward being reversed or corrected	rw_example_0001	Conditional	-	Must exist; not self	Backend	Internal	Link reversals	Proposed
Rewards	wallet_transaction_id	string	Wallet entry for the credit	wt_example_0001	Conditional	-	Must exist in Wallet_Transactions	Backend	Internal	Reconcile rewards and wallet	Proposed
Rewards	reason_category	enum	Reason for reversal, rejection or correction	operational_error	Conditional	To be agreed with Operations	No free text	Backend	Internal	Explain reversals	To Be Verified
Wallet_Transactions	transaction_id	string	Key of one ledger entry	wt_example_0001	Yes	-	Unique; immutable; server-generated	Backend	Internal	Deduplication and reconciliation	To Be Verified
Wallet_Transactions	idempotency_key	string	Key making repeated requests create one entry	idem_example_01	No	-	Unique per request	Backend	Internal	Prevent double credits	Proposed
Wallet_Transactions	user_id	string	Wallet owner	usr_example_0001	Yes	-	Must exist in Users	Backend	Internal	Join to Users	To Be Verified
Wallet_Transactions	created_at	timestamp (UTC)	When the entry was created	2025-01-15T11:02:31 Z	Yes	-	Not in the future	Backend	Internal	Ledger timing	Proposed
Wallet_Transactions	completed_at	timestamp (UTC)	When the entry became final	2025-01-15T11:02:31 Z	Conditional	-	Required when completed	Backend	Internal	Settlement time	Proposed
Wallet_Transactions	transaction_type	enum	Kind of entry	reward_credit	Yes	reward_credit, redemption, withdrawal_debit, exchange_debit, exchange_credit, reversal, adjustment	In approved list	Backend	Internal	Split ledger by type	Proposed
Wallet_Transactions	direction	enum	Credit or debit	credit	Yes	credit, debit	Agrees with transaction_type	Backend	Internal	Sign of the entry	Proposed
Wallet_Transactions	amount	decimal	Size of the entry, always positive	10.00	Yes	-	Greater than 0	Backend	Restricted	Volumes and balances	Proposed
Wallet_Transactions	unit	enum	Unit of the amount	unit_example	Yes	Real VELOOP units	One unit per entry	Backend	Internal	Prevents mixed-unit sums	To Be Verified
Wallet_Transactions	status	enum	State of the entry	completed	Yes	created, completed, failed, reversed	In approved list	Backend	Internal	Ledger state mix	Proposed
Wallet_Transactions	source_feature	enum	Feature that caused the entry	task	Yes	Real VELOOP features	In approved list	Backend	Internal	Activity by feature	Proposed
Wallet_Transactions	reference_id	string	Record that caused the entry	rw_example_0001	Yes	-	Must exist in referenced table	Backend	Internal	Reconciliation	Proposed
Wallet_Transactions	related_transaction_id	string	Entry being reversed or adjusted	wt_example_0000	Conditional	-	Required for reversal and adjustment; not self	Backend	Internal	Link reversals	Proposed
Wallet_Transactions	exchange_group_id	string	Pairs the two entries of an exchange	ex_example_01	Conditional	-	One debit and one credit	Backend	Internal	Exchange integrity	Proposed
Wallet_Transactions	balance_after	decimal	Balance in that unit after the entry	120.00	No	-	Not negative; matches previous balance	Backend	Restricted	Balance reconciliation	To Be Verified
Withdrawals	withdrawal_id	string	Key of one withdrawal request	wd_example_0001	Yes	-	Unique; immutable	Backend	Internal	Grain of the table	To Be Verified
Withdrawals	user_id	string	Requester	usr_example_0001	Yes	-	Must exist in Users	Backend	Internal	Join to Users	To Be Verified

Entity	Field Name	Data Type	Definition	Example	Required	Allowed Values	Validation	Source	Sensitivity	Analytical Use	Status
Withdrawals	requested_amount	decimal	Amount requested	50.00	Yes	-	Greater than 0	Backend	Restricted	Withdrawal volume	To Be Verified
Withdrawals	requested_unit	enum	Unit of the request	unit_example	Yes	Real VELOOP units	In approved list	Backend	Internal	Prevents mixed-unit sums	To Be Verified
Withdrawals	method	enum	Payout method	method_example	Yes	Real VELOOP methods	In approved list	Backend	Restricted	Method distribution	To Be Verified
Withdrawals	requested_at	timestamp (UTC)	When requested	2025-02-01T12:00:00Z	Yes	-	Not in the future	Backend	Internal	Request timing	Proposed
Withdrawals	processing_at	timestamp (UTC)	When processing began	2025-02-01T14:00:00Z	Conditional	-	Not before requested_at	Backend	Internal	Processing time	Proposed
Withdrawals	final_at	timestamp (UTC)	When a terminal status was reached	2025-02-02T10:00:00Z	Conditional	-	Required for terminal statuses	Backend	Internal	Processing time	Proposed
Withdrawals	status	enum	Current state	completed	Yes	requested, processing, completed, rejected, reversed (cancelled if allowed)	In approved list	Backend	Internal	Lifecycle counts	Proposed
Withdrawals	reason_category	enum	Reason for rejection or reversal	insufficient_balance	Conditional	insufficient_balance, information_incomplete, method_unavailable, policy_review, provider_failure, user_request, operational_error, other	Required when rejected or reversed	Backend	Internal	Explain rejections	Proposed
Withdrawals	completed_amount	decimal	Amount actually paid	50.00	Conditional	-	Required when completed	Backend	Restricted	Paid volume	To Be Verified
Withdrawals	wallet_transaction_id	string	Wallet entry for this withdrawal	wt_example_0007	Conditional	-	Exactly one active debit	Backend	Internal	Reconcile with wallet	Proposed
Withdrawals	reviewer_type	enum	Automated or manual handling	manual	No	automated, manual, unknown	In approved list	Backend	Internal	Interpret processing time	To Be Verified
Withdrawals	payout_provider_reference_id	string	Payout partner's reference	pp_example_01	No	-	Unique per provider	Provider	Restricted	Match external payouts	To Be Verified
Withdrawals	reversed_at	timestamp (UTC)	When a completed payout was reversed	2025-02-05T10:00:00Z	Conditional	-	Required when reversed	Backend	Internal	Reversal timing	Proposed
Referrals	referral_id	string	Key of one referral relationship	ref_example_0001	Yes	-	Unique; immutable	Backend	Internal	Grain of the table	Proposed
Referrals	referring_user_id	string	Inviting user	usr_example_0001	Yes	-	Must exist in Users	Backend	Internal	Referrer analysis	To Be Verified
Referrals	referred_user_id	string	Invited user once registered	usr_example_0002	Conditional	-	Must differ from referring_user_id	Backend	Internal	Referred user analysis	To Be Verified
Referrals	referral_code	string	Code or link token used	REF_EXAMPLE1	No	-	Unique per referring user	Backend	Internal	Trace the source	To Be Verified
Referrals	created_at	timestamp (UTC)	When the referral was created	2025-01-20T09:00:00Z	Yes	-	Not in the future	Backend	Internal	Funnel start	Proposed
Referrals	registered_at	timestamp (UTC)	When the referred user registered	2025-01-21T09:00:00Z	Conditional	-	Not before created_at	Backend	Internal	Registration step	Proposed
Referrals	activated_at	timestamp (UTC)	When the activation rule was met	2025-01-22T09:00:00Z	Conditional	-	Not before registered_at	Backend	Internal	Activation step	To Be Verified
Referrals	converted_at	timestamp (UTC)	When the success condition was met	2025-01-25T09:00:00Z	Conditional	-	Not before activated_at	Backend	Internal	Conversion step	Proposed
Referrals	status	enum	Current referral stage	activated	Yes	created, registered, activated, converted, closed	In approved list	Backend	Internal	Funnel state	Proposed
Referrals	source_channel	enum	How the referral was shared	link	No	link, code, other	In approved list	Backend	Internal	Channel effectiveness	Proposed
Referrals	activation_rule_version	string	Version of the activation rule applied	v1	No	-	Matches change log	Backend	Internal	Explain results across rule changes	Proposed
Referrals	referral_reward_id	string	Reward given for the referral	rw_example_0009	No	-	Must exist in Rewards	Backend	Internal	Reward cost of referrals	To Be Verified
Referrals	closed_reason_category	enum	Why closed without converting	other	Conditional	To be agreed	No free text	Backend	Internal	Explain lost referrals	To Be Verified

Entity	Field Name	Data Type	Definition	Example	Required	Allowed Values	Validation	Source	Sensitivity	Analytical Use	Status
Feature_Events	event_id	string	Key of one event occurrence	ev_example_0001	Yes	-	Unique; retries reuse it	Frontend / Backend	Internal	Deduplication	Proposed
Feature_Events	event_name	string	Name from the event catalogue	task_completed	Yes	Event catalogue names	Must be in catalogue	Frontend / Backend	Internal	Identify the event	Proposed
Feature_Events	user_id	string	User who triggered the event	usr_example_0001	Yes	-	Must exist in Users	Frontend / Backend	Internal	Join to Users	To Be Verified
Feature_Events	event_timestamp	timestamp (UTC)	When the event happened	2025-01-15T11:02:30 Z	Yes	-	Not in the future	Frontend / Backend	Internal	Ordering and daily metrics	Proposed
Feature_Events	received_timestamp	timestamp (UTC)	When the pipeline received it	2025-01-15T11:02:32 Z	Yes	-	Not before event_timestamp by more than clock tolerance (to be agreed)	Pipeline	Internal	Detect delay and clock drift	Proposed
Feature_Events	schema_version	string	Version of the event definition	1	Yes	-	Known version	Frontend / Backend	Internal	Handle definition changes	Proposed
Feature_Events	platform	enum	Platform that emitted it	android	Yes	android, ios, web	In approved list	Frontend / Backend	Internal	Platform split	Proposed
Feature_Events	app_version	string	App version that emitted it	1.4.2	Yes	-	Version pattern	Frontend	Internal	Version problems	Proposed
Feature_Events	source	enum	Emitter type	backend	Yes	frontend, backend, provider	In approved list	Frontend / Backend	Internal	Trust in the event	Proposed
Feature_Events	session_id	string	Visit key, if session tracking exists	sess_example_01	No	-	Unique per session	Frontend	Internal	Session analysis	Proposed
Feature_Events	feature	enum	Product feature of the event	spin	Yes	Real VELOOP features	In approved list	Frontend / Backend	Internal	Feature usage	Proposed
Feature_Events	properties	json	Event-specific properties from the catalogue	{"task_id": "task_example_01"}	Yes	Per event definition	Valid JSON; required properties present	Frontend / Backend	Internal	Carry event details	Proposed
Support_Tickets	ticket_id	string	Key of one support request	tk_example_0001	Yes	-	Unique	Backend	Internal	Grain of the table	To Be Verified
Support_Tickets	user_id	string	Requesting user	usr_example_0001	Yes	-	Must exist in Users	Backend	Internal	Join to Users	To Be Verified
Support_Tickets	issue_category	enum	Category of the issue	withdrawal	Yes	To be agreed	In list; no free text	Backend	Internal	Issue mix	To Be Verified
Support_Tickets	created_at	timestamp (UTC)	When the request was created	2025-02-03T10:00:00 Z	Yes	-	Not in the future	Backend	Internal	Support demand	To Be Verified
Support_Tickets	resolved_at	timestamp (UTC)	When it was resolved	2025-02-03T15:00:00 Z	Conditional	-	Not before created_at	Backend	Internal	Resolution time	To Be Verified
Support_Tickets	ticket_status	enum	State of the request	resolved	Yes	open, resolved (to be agreed)	In approved list	Backend	Internal	Open workload	To Be Verified